## Question

The artificial nitrogen fixation process is one of the lifelines of mankind, and the industrial preparation of ammonia is the most important reaction in it.

1. At $700K$, in a $1.0\ m^3$ container, $1.0\ mol\ \mathrm{N_2}$ and $3.0\ mol\ \mathrm{H_2}$ react to produce ammonia. The thermodynamic data are shown in Table 1:

| - | $N_2(g)$ | $H_2(g)$ | $NH_3(g)$ |
|---|---|---|---|
| $\Delta_f H_m^\circ / kJ \cdot mol^{-1}$ | -- | -- | -45.9 |
| $S_m^\circ / J \cdot mol^{-1} \cdot K^{-1}$ | 191.6 | 130.7 | 192.8 |

2. Cool the ammonia produced after the system in question 1 reaches equilibrium from $700\ K$ to $50\ K$ (always maintaining a pressure of $1\ bar$). The relevant thermodynamic data of ammonia are shown in Table 2:

| | $\Delta_{vap} H_m^\circ$ | $\Delta_{vap} S_m^\circ$ | $\Delta_{fus} H_m^\circ$ | $\Delta_{fus} S_m^\circ$ | $C_{p,m}(g)$ | $C_{p,m}(l)$ | $C_{p,m}(s)$ |
|---|---|---|---|---|---|---|---|
| $NH_3(g)$ | 23.5 | 98.3 | 5.66 | 32.0 | 35.1 | 78.5 | 78.5 (approximate) |

The unit of heat capacity in the table is $J \cdot mol^{-1} K^{-1}$, the unit of enthalpy change is $kJ \cdot mol^{-1}$, and the unit of entropy change is $J \cdot mol^{-1} K^{-1}$.

Which of the following options is correct:

**A.** In question 1, the $K^\circ$ for ammonia synthesis is $4.58 \times 10^{-4}$

**B.** The conversion rate of ammonia synthesis in question 1 is $0.54\%$

**C.** Due to the presence of hydrogen bonds, solid ammonia and water exhibit similar properties upon melting.

**D.** In problem 2, the enthalpy change of the entire process is $\Delta H = -60.2 kJ$

**E.** The melting point of ammonia decreases as pressure increases.

**F.** All of the above options are incorrect.

## Answer

Correct Answer: **F**

## Detailed Explanation

Problem 1:

$$\Delta_r G_m^\circ(700K) = \Delta_r H_m^\circ(700K) - T \cdot \Delta S_m^\circ(700K) = -45.9kJ \cdot mol^{-1} \times 2 - 700K \times (192.8 \times 2 - 191.6 \times 1 - 130.7 \times 3) J \cdot mol^{-1} \cdot K^{-1} = 46.87kJ \cdot mol^{-1}$$

CHECKPOINT 1 PTS

$$\Delta_r G_m^\circ(700K) = 46.87kJ \cdot mol^{-1}$$

$$\Delta_r G_m^\circ = -RTln(K^\circ)$$

$$-\Delta_r G_m^\circ/(RT) = -46.87kJ \cdot mol^{-1}/(8.314J \cdot mol^{-1} \cdot K^{-1} \times 700K) \approx -8.054$$

$$K^\circ = e^{-8.054} \approx 3.180 \times 10^{-4}$$

CHECKPOINT 1 PTS

$$K^\circ = 3.180 \times 10^{-4}$$

Equation: $\mathrm{N_2(g) + 3H_2(g) = 2NH_3(g)}$

According to $pV = nRT$, the initial $\mathrm{N_2(g)}$ is calculated to be $0.05820 bar$, and $\mathrm{H_2(g)}$ is $0.1746 bar$

Let the amount of $\mathrm{NH_3(g)}$ produced be $2x$, then the amount of $\mathrm{N_2(g)}$ is $0.05820 - x$, and the amount of $\mathrm{H_2(g)}$ is $0.1746 - 3x$, in units of $bar$

List the equation

$$3.180 \times 10^{-4} = \frac{4x^2 \cdot (p^\circ)^2}{(0.05820 - x)(0.1746 - 3x)^3}$$

Solving for $x$ yields $x = 1.56 \times 10^{-4}$

**CHECKPOINT** 1 PTS

$x = 1.56 \times 10^{-4}$

The conversion rate is $0.27\%$

**CHECKPOINT** 1 PTS

The conversion rate is $0.27\%$

Due to the presence of hydrogen bonds, solid ammonia and water exhibit different properties upon melting, because hydrogen atoms in solid ammonia participate in hydrogen bond formation, and a large number of hydrogen bonds make the structure more compact and denser.

**CHECKPOINT** 1 PTS

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Problem 2:

Now calculate the melting point and boiling point based on the enthalpy change and entropy change, construct a cycle, and finally calculate the enthalpy change of the entire process.

$$T_b = \frac{\Delta_{vap} H_m^\circ}{\Delta_{vap} S_m^\circ} = \frac{23.5 kJ \cdot mol^{-1}}{98.3 J \cdot mol^{-1} \cdot K^{-1}} = 239 K$$

**CHECKPOINT** 1 PTS

$$T_b = \frac{\Delta_{vap}H_m^\circ}{\Delta_{vap}S_m^\circ} = \frac{23.5kJ \cdot mol^{-1}}{98.3J \cdot mol^{-1} \cdot K^{-1}} = 239K$$

$$T_f = \frac{\Delta_{fus}H_m^\circ}{\Delta_{fus}S_m^\circ} = \frac{5.66kJ \cdot mol^{-1}}{32.0J \cdot mol^{-1} \cdot K^{-1}} = 177K$$

**CHECKPOINT** 1 PTS

$$T_f = \frac{\Delta_{fus}H_m^\circ}{\Delta_{fus}S_m^\circ} = \frac{5.66kJ \cdot mol^{-1}}{32.0J \cdot mol^{-1} \cdot K^{-1}} = 177K$$

The entire process can be expressed as

$$\Delta H = \Delta H_1(5.4mmol, g, 700K \to 239K, 1bar) + \Delta H_2(5.4mmol, g \to l, 239K, 1bar) + \Delta H_3(5.4mmol, l, 239K \to 177K, 1bar) + \Delta H_4(5.4mmol, l \to s, 177K,$$

$$\Delta H = 5.4 \times 10^{-3} mol \times [(239-700)K \times 35.1J \cdot mol^{-1} \cdot K^{-1} + (177-239)K \times 78.5J \cdot mol^{-1} \cdot K^{-1} + (50-177)K \times 78.5J \cdot mol^{-1} \cdot K^{-1} - 23.5kJ \cdot mol^{-1} - 5.6$$

**CHECKPOINT** 1 PTS

$$\Delta H = -320J$$

According to the Clapeyron equation:

$$\frac{\Delta P}{\Delta T} = \frac{\Delta H}{T\Delta V}$$

For the solidification process

$$\Delta H < 0,\ T > 0,\ \Delta V < 0$$

Therefore

$$\frac{\Delta P}{\Delta T} > 0$$

The melting point of ammonia increases with increasing pressure.

**CHECKPOINT** 1 PTS

The melting point of ammonia increases with increasing pressure

The final result is F